

Maisha Maimuna

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Arlington, Texas - 76013, USA

RESEARCH INTEREST

My research interest lies in human-centered computing (HCC) and human-robot interaction (HRI) with a focus on assistive and collaborative technologies. I have experience working with machine learning models and large language models (LLMs), including prompt design and the development of multimodal feedback system in simulation environment to support navigation for visually impaired individuals.

EDUCATION

Ph.D. in Computer Science University of Texas at Arlington	<i>Jan 2024 – Present</i> Arlington, USA
• CGPA: 4.00	
B.Sc. in Computer Science and Engineering Chittagong University of Engineering and Technology	<i>August 2019</i> Chattogram, Bangladesh
• CGPA: 3.58	

WORK EXPERIENCE

Graduate Researcher - Heracleia Human-Centered Computing Lab University of Texas at Arlington	<i>May 2024 – Present</i> Arlington, USA
Graduate Teaching Assistant University of Texas at Arlington	<i>Jan 2024 – Present</i> Arlington, USA
Lecturer - Computer Science and Engineering Southern University Bangladesh	<i>Jan 2020 – Jan 2023</i> Chattogram, Bangladesh

PUBLICATIONS

Conference Papers

- Karim, E., Maimuna, M., Roy, A., Bin Farukee, M., Nale, G., & Makedon, F. (2025, June). An Adaptive Robotic Framework for Assisting People with Paralysis in Daily Living Activities. *In Proceedings of the 18th ACM International Conference on PErvasive Technologies Related to Assistive Environments* (pp. 419-423).

Book Chapters

- Maimuna, M., Hossain, S. M. M., & Deb, K. (2022, October). Masked Face Inpainting Using Generative Adversarial Network Based Architecture. *In International Conference on Intelligent Computing & Optimization* (pp. 265-274). Cham: Springer International Publishing.
- Maimuna, M., Rahman, N., Ahmed, R., & Arefin, M. S. (2021, December). Data mining for software engineering: a survey. *In International Conference on Intelligent Computing & Optimization* (pp. 905-916). Cham: Springer International Publishing.
- Rahman, N., Maimuna, M., Begum, A., Ahmed, M. R., & Arefin, M. S. (2021). A survey of data mining techniques in the field of cyborg mining. *In Soft Computing for Security Applications: Proceedings of ICSCS 2021* (pp. 781-797). Singapore: Springer Singapore.
- Chakraborty, A., Das, U. K., Sikder, J., Maimuna, M., & Sarek, K. I. (2022, October). Content based email spam classifier as a web application using naïve bayes classifier. *In International Conference on Intelligent Computing & Optimization* (pp. 389-398). Cham: Springer International Publishing.
 - Best Reviewed Paper Award

Preprints

- Maimuna, M., Farukee, M. B., Nikanfar, S., Siddiqua, M., Roy, A., & Makedon, F. (2025). NavVI: A Telerobotic Simulation with Multimodal Feedback for Visually Impaired Navigation in Warehouse Environments. *arXiv preprint arXiv:2507.15072*.

Manuscripts Under Review

- Maimuna, M., Farukee, M. B., Nikanfar, S., Siddiqua, & Makedon, F. NavVI: Designing and Evaluating Multimodal Guidance to Assist People with Vision Impairments in Telerobotic Warehouse Navigation. *In review*.
- Karim, E., Maimuna, M., Bin Farukee, & Makedon, F. Mindful Assistant: Personalized Assistance in Daily Living Activities for Individuals with Paralysis via Multimodal Cognitive Fatigue Assessment. *In review*.

AWARDS & HONORS

- **CSE TA of the Month** November, 2025
The University of Texas at Arlington
- **Best Reviewed Paper Award** October, 2022
ICO 2022 Conference
- **Phi Kappa Phi - Academic Honor** April, 2026
The University of Texas at Arlington

SERVICE & OUTREACH

Program Committee Member

- The PErvasive Technologies Related to Assistive Environments (PETRA) conference 2025, 2026

Conference Advertising & Editorial Board, and Reviewer

- PETRA conference 2025, 2026

Workshop Chair

- GenAssist 2026: Generative AI and Large Language Models in Assistive Technologies, PETRA conference 2026

TEACHING

Teaching Assistant - University of Texas at Arlington

- DASC-5303-Data Science Project Management Spring 2024
- CSE-5360-Artificial Intelligence Summer 2024 - Spring 2025
- CSE-6363-Machine Learning Summer 2025
- CSE-6331-ADV Topics in Database Systems Fall 2025
- CSE-5306-Distributed Systems Spring 2026

Lecturer - Southern University Bangladesh

- Digital Logic Design Spring 2020 - Fall 2022
- Numerical Methods Spring 2020 - Fall 2022
- Parallel & Distributed Processing Spring 2020
- Robotics & Computer Vision Spring 2020 - Fall 2022
- Data Structure Spring 2021
- Computer Basics & Programming Concepts Fall 2021
- Digital Image Processing Spring 2022, Fall 2022
- Algorithm Design & Analysis Spring 2022, Fall 2022

POSTER PRESENTATIONS

- An Adaptive Robotic Assistance System for Enhancing Daily Living Activities in Individuals with Paralysis *Poster Presentation at Innovation day 2025, The University of Texas at Arlington*
- Exploring the Feasibility of Robotic Teleoperation for Visually Impaired Individuals *Poster Presentation at Student Computing Research Festival (SCRF) 2025., The University of Texas at Arlington*

CERTIFICATIONS

- Introduction to Large Language Models: Linked in Learning June 2024
- Deep Learning A-Z, 2024: Udemy June 2024
- Generative AI with Large Language Models: Coursera July 2024